Lab Assignment 03

BRAC UNIVERSITY

Inspiring Excellence

Course Code:	**CSE111**
Course Title:	**Programming Language II**
Topic:	**Method Overloading, Constructor & Overloading**
Number of Tasks:	**12 (Classwork: 06, Homework: 06)**

[Submit all the Coding Tasks (Homework: Task 1 to 4) in the Google Form shared on buX before the next lab. Submit the Tracing Tasks (Homework: Task 5 to 6) handwritten to your Lab Instructors at the beginning of the lab]

[You are not allowed to change the driver codes of any of the tasks]

CLASSWORK

Task 1

Design the **Shape** class with necessary properties to produce the given output for the provided driver code.

Driver Class	Output

`public class ShapeTester{`
`  public static void main(String args []){`
`  Shape circle = new Shape();`
`  Shape triangle = new Shape();`
`  Shape rectangle = new Shape();`
 	
`  circle.setParameters("Circle", 5);`
`  triangle.setParameters("Triangle", 4, 7);`
`  rectangle.setParameters("Rectangle", 2.4, 4.4);`
 	
`  System.out.println(circle.details());`
`  System.out.println("1---------------");`
`  System.out.println(triangle.details());`
`  System.out.println("2---------------");`
`  System.out.println(rectangle.details());`
`  }`
`}`
	`Shape Name: Circle`
	`Area: 78.54`
	`1---------------`
	`Shape Name: Triangle`
	`Area: 14.0`
	`2---------------`
	`Shape Name: Rectangle`
	`Area: 10.56`

Task 2

Design the **Student** class in such a way that it produces the following output.

Driver Code	Expected Output

`public class StudentTester{`
`  public static void main(String[] args){`
`  Student s1 = new Student("Harry", "CSE");`
`  System.out.println(s1.name);`
`  s1.updateName("Harry Potter");`
`  System.out.println(s1.name);`
`  System.out.println(s1.prog);`
`  s1.updateProgram("CS");`
`  String var = s1.accessProgram();`
`  System.out.println(var);`
`  }`
`}`
	`Harry`
	`Harry Potter`
	`CSE`
	`CS`

Task 3

Design the **Shape2D** class in such a way that it produces the following output.

Driver Code	Expected Output

`public class Shape2DTester {`
`  public static void main(String[] args) {`
`  Shape2D sq = new Shape2D();`
`  System.out.println("---------1----------");`
`  sq.area();`
`  System.out.println("---------2----------");`
`  Shape2D rectangle = new Shape2D(5, 6);`
`  System.out.println("---------3----------");`
`  rectangle.area();`
`  System.out.println("---------4----------");`
`  Shape2D tri1 = new Shape2D(5, 6, "Triangle");`
`  System.out.println("---------5----------");`
`  tri1.area();`
`  System.out.println("---------6----------");`
`  Shape2D tri2 = new Shape2D(5, 6, 7);`
`  System.out.println("---------7----------");`
`  tri2.area();`
`  }`
`}`
	`A Square has been created with length: 5`
	`---------1----------`
	`The area of the Square is: 25.00`
	`---------2----------`
	`A Rectangle has been created with length: 5 and breath:  6`
	`---------3----------`
	`The area of the Rectangle is: 30.00`
	`---------4----------`
	`A Triangle has been created with height: 5 and base:  6`
	`---------5----------`
	`The area of the Triangle is: 15.00`
	`---------6----------`
	`A Triangle has been created with the following sides: 5, 6, 7`
	`---------7----------`
	`The area of the Triangle is: 14.70`

Task 4

Design the **Library** class with the necessary properties so that the given output is produced for the provided driver code.

Driver Code	Output

`public class Tester5{`
`  public static void main(String[] args) {`
`  Library a1 = new Library(3);`
`  System.out.println("1----------");`
`  a1.addBook("Ice");`
`  System.out.println("2----------");`
`  a1.printDetail();`
`  System.out.println("3----------");`
`  a1.addBook("Emma");`
`  a1.addBook("Wings");`
`  a1.addBook("Next");`
`  System.out.println("4----------");`
`  a1.printDetail();`
`  System.out.println("5----------");`
`  Library a2 = new Library(4);`
`  System.out.println("6----------");`
`  a2.addBook("Onnobhubon");`
`  a2.addBook("Ami");`
`  System.out.println("7----------");`
`  a2.printDetail();`
`  System.out.println("8----------");`
`  a2.addBook("Deyal");`
`  a2.addBook("Himu");`
`  a2.addBook("Megher Upor Bari");`
`  System.out.println("9----------");`
`  a2.printDetail();`
`  }`
`}`
	`A library has been created with capacity 3`
	`1----------`
	`Book 'Ice' added to the library`
	`2----------`
	`Maximum Capacity: 3`
	`Total Books: 1`
	`Book list:`
	`Ice`
	`3----------`
	`Book 'Emma' added to the library`
	`Book 'Wings' added to the library`
	`Exceeds maximum capacity. You can't add more than 3 books`
	`4----------`
	`Maximum Capacity: 3`
	`Total Books: 3`
	`Book list:`
	`Ice`
	`Emma`
	`Wings`
	`5----------`
	`A library has been created with capacity 4`
	`6----------`
	`Book 'Onnobhubon' added to the library`
	`Book 'Ami' added to the library`
	`7----------`
	`Maximum Capacity: 4`
	`Total Books: 2`
	`Book list:`
	`Onnobhubon`
	`Ami`
	`8----------`
	`Book 'Deyal' added to the library`
	`Book 'Himu' added to the library`
	`Exceeds maximum capacity. You can't add more than 4 books`
	`9----------`
	`Maximum Capacity: 4`
	`Total Books: 4`
	`Book list:`
	`Onnobhubon`
	`Ami`
	`Deyal`
	`Himu`

Task 5

Design the **TaxiLagbe** class with necessary properties to produce the given output for the provided driver code.

Driver Code

```
public class TaxiTester{
 public static void main(String[] args) {
   TaxiLagbe taxi1 = new TaxiLagbe("1010-01","Dhaka");
   System.out.println("1-------------");
   taxi1.printDetails();
   System.out.println("2-------------");
   taxi1.addPassenger("Wilson", 105);
   System.out.println("3-------------");
   taxi1.printDetails();
   System.out.println("4-------------");
   taxi1.addPassenger("Walker", 100, "Wood", 200);
   System.out.println("5-------------");
   taxi1.printDetails();
   System.out.println("6-------------");
   taxi1.addPassenger("Karen", 200);
   taxi1.addPassenger("Donald", 130);
   System.out.println("7-------------");
   taxi1.printDetails();
   System.out.println("8-------------");
   TaxiLagbe taxi2 = new TaxiLagbe("1010-02");
   System.out.println("9-------------");
   taxi2.addPassenger("Don", 115, "Parker", 215);
   System.out.println("10-------------");
   taxi2.printDetails();
 }
}
```

Output

```
A new taxi has been created
1-------------
Taxi number: 1010-01
This taxi can cover Dhaka area
Total Passenger: 0
Passenger List:
Total collected fare: 0 Taka
2-------------
Dear Wilson! Welcome to TaxiLagbe
3-------------
Taxi number: 1010-01
This taxi can cover Dhaka area
Total Passenger: 1
Passenger List:
Wilson
Total collected fare: 105 Taka
4-------------
Dear Walker! Welcome to TaxiLagbe
Dear Wood! Welcome to TaxiLagbe
5-------------
Taxi number: 1010-01
This taxi can cover Dhaka area
Total Passenger: 3
Passenger List:
Wilson Walker Wood
Total collected fare: 405 Taka
6-------------
Dear Karen! Welcome to TaxiLagbe
Taxi Full! No more passengers can be
added
7-------------
Taxi number: 1010-01
This taxi can cover Dhaka area
Total Passenger: 4
Passenger List:
Wilson Walker Wood Karen
Total collected fare: 605 Taka
8-------------
A new taxi has been created
9-------------
Dear Don! Welcome to TaxiLagbe
Dear Parker! Welcome to TaxiLagbe
10-------------
Taxi number: 1010-02
This taxi can cover Khulna area
Total Passenger: 2
Passenger List:
Don Parker
Total collected fare: 330 Taka
```

Task 6

```
public class A{
  public int temp = 3, sum = 9, y = 4, x = 0;
  public A(){
    int sum = 7;
    y = temp - 5;
    sum = temp + 2;
    temp -= 2;
    this.x = sum + temp + y;
  }
  public A(int y, int temp){
    y = temp - 1 + x;
    sum = temp + 2 -x;
    temp-=2;
  }
  public void methodB(int m, int [] n){
    int x = 0;
    y = y + m + this.methodB(x,m);
    x = this.x + 2 + (++n[0]);
    sum = sum + x + y;
    n[0] = sum + 2;
    System.out.println(n[0] + " " + y+ " " + sum);
  }
  public int methodB(int m, int n){
    int [] y = {10};
    this.y = y[0] + this.y + m;
    x = this.y + 2 + temp - n;
    sum = x + y[0] + this.sum;
    System.out.println(y[0]+ " "+ temp + " " + sum);
    return y[0];
  }
}
```

Driver Code	Output		

`public class Tester6 {`
`  public static void main(String args[]){`
`  int[] x = {35};`
`  A a1;`
`  a1 = new A();`
`  A a2 = new A(-5,-7);`
`  a1.methodB(1, x);`
`  a2.methodB(1, x);`
`  }`
`}`			

HOMEWORK

Task 1

Design the **Toy** class in such a way that it produces the following output

Driver Code	Expected Output

`public class ToyTester{`
`  public static void main(String[] args){`
`  Toy t1 = new Toy("Car", 230);`
`  System.out.println("1==========");`
`  t1.updatePrice(250);`
`  System.out.println("2==========");`
`  System.out.println(t1.name);`
`  t1.showPrice();`
`  System.out.println("3==========");`
`  Toy t2 = new Toy("Robot", 450);`
`  System.out.println("4==========");`
`  t2.updateName("Autobot");`
`  System.out.println("5==========");`
`  System.out.println(t2.name);`
`  t2.showPrice();`
`  }`
`}`
	`A new toy has been made!`
	`1==========`
	`2==========`
	`Car`
	`price: 250 Taka`
	`3==========`
	`A new toy has been made!`
	`4==========`
	`Changing old name: Robot`
	`new name: Autobot`
	`5==========`
	`Autobot`
	`price: 450 Taka`

Task 2

Complete the following **Cart** class to generate the given output from the tester code:

- A cart will have a cart number which will be assigned in ***create_cart()*** method.
- Each cart can hold up to 3 items (at max).
- Each cart must have two arrays to store items and their respective prices.
- The items inside a cart will be added in ***addItem()*** method only if the cart items do not exceed 3.
- The ***giveDiscount()*** method saves the discount given to that cart object and updates the price accordingly.

Driver Code	Output

`public class CartTester{`
`  public static void main(String [] args){`
`  Cart c1 = new Cart ();`
`  Cart c2 = new Cart ();`
`  Cart c3 = new Cart ();`
 	
`  c1.create_cart(1);`
`  c2.create_cart(2);`
`  c3.create_cart(3);`
`  System.out.println("====1====");`
`  c1.addItem("Table", 3900.5);`
`  c1.addItem("Chair", 1400.76);`
`  c1.addItem(5400.87, "Television");`
`  c1.addItem(5000.0, "Refrigerator");`
 	
`  System.out.println("====2====");`
`  c2.addItem("Stove",439.90);`
 	
`  System.out.println("====3====");`
`  c3.addItem("Chair",1400.5);`
`  c3.addItem(3400.0, "Chair");`
 	
`  System.out.println("====4====");`
`  c1.cartDetails();`
 	
`  System.out.println("====5====");`
`  c2.cartDetails();`
 	
`  System.out.println("====6====");`
`  c3.cartDetails();`
`  c1.giveDiscount(10);`
 	
`  System.out.println("====7====");`
`  c1.cartDetails();`
`  }`
`}`
	`====1====`
	`Table added to cart 1.`
	`You have 1 item(s) in your cart now.`
	`Chair added to cart 1.`
	`You have 2 item(s) in your cart now.`
	`Television added to cart 1.`
	`You have 3 item(s) in your cart now.`
	`You already have 3 items on your cart`
	`====2====`
	`Stove added to cart 2.`
	`You have 1 item(s) in your cart now.`
	`====3====`
	`Chair added to cart 3.`
	`You have 1 item(s) in your cart now.`
	`Chair added to cart 3.`
	`You have 2 item(s) in your cart now.`
	`====4====`
	`Your cart(c1) :`
	`Table - 3900.5`
	`Chair - 1400.76`
	`Television - 5400.87`
	`Discount Applied: 0.0%`
	`Total price: 10702.130000000001`
	`====5====`
	`Your cart(c2) :`
	`Stove - 439.9`
	`Discount Applied: 0.0%`
	`Total price: 439.9`
	`====6====`
	`Your cart(c3) :`
	`Chair - 1400.5`
	`Chair - 3400.0`
	`Discount Applied: 0.0%`
	`Total price: 4800.5`
	`====7====`
	`Your cart(c1) :`
	`Table - 3900.5`
	`Chair - 1400.76`
	`Television - 5400.87`
	`Discount Applied: 10.0%`
	`Total price: 9631.917000000001`

Task 3

Design the **Reader** class in such a way so that the following code provides the expected output.

- A reader will have a name, capacity to read and an array of books they are reading.
- The initial capacity of a reader will be 2. The initial name will be "New user".

Driver Code

```
public class Reader_tester {
  public static void main(String[] args){

    System.out.println("1 ==========");
    Reader r1 = new Reader("Messi");
    Reader r2 = new Reader("Ronaldo", 3);
    System.out.println("2 ==========");
    r1.readerInfo();
    System.out.println("3 ==========");
    r2.addBook("Java");
    r2.addBook("Python");
    r2.addBook("C++");
    r2.readerInfo();
    System.out.println("4 ==========");
    r1.addBook("C#");
    r1.addBook("Rust");
    r1.addBook("GoLang");
    System.out.println("5 ==========");
    r2.addBook("Python");
    System.out.println("6 ==========");
    r1.readerInfo();
    System.out.println("7 ==========");
    r1.updateCapacity(3);
    System.out.println("8 ==========");
    r1.addBook("GoLang");
    System.out.println("9 ==========");
    r1.readerInfo();

  }
}
```

Expected Output

```
1 ==========
A new reader is created!
A new reader is created!
2 ==========
Name: Messi
Capacity: 2
Books:
No books added yet
3 ==========
Name: Ronaldo
Capacity: 3
Books:
Book 1: Java
Book 2: Python
Book 3: C++
4 ==========
No more capacity
5 ==========
No more capacity
6 ==========
Name: Messi
Capacity: 2
Books:
Book 1: C#
Book 2: Rust
7 ==========
Capacity has changed to 3
8 ==========
9 ==========
Name: Messi
Capacity: 3
Books:
Book 1: C#
Book 2: Rust
Book 3: GoLang
```

Task 4

You are building a tracker system that will keep track of a person's income and expenses.

- When the ***createTracker()*** method is invoked it sets the balance to 1.0 taka.
- The ***info()*** method **returns** a String with the trackers information.
- If the total balance becomes 0 after the ***expense()*** method is called it prints "You're broke!" . Again if the available balance is less than the expense it prints "Not enough balance.". Otherwise the method prints "Balance updated" after updating the balance.
- The last expense and income history can be seen by using the ***history()*** method.

Driver Code

```
public class Tester4{
 public static void main(String[] args) {
    MoneyTracker tr1 = new MoneyTracker();
    System.out.println(tr1.info());
    tr1.createTracker("John");
    System.out.println("1 ==========");
    System.out.println(tr1.info());
    System.out.println("2 ==========");
    tr1.income(1000);
    System.out.println(tr1.info());
    System.out.println("3 ==========");
    tr1.expense(800);
    tr1.expense(100);
    System.out.println(tr1.info());
    System.out.println("4 ==========");
    tr1.showHistory();
    System.out.println("5 ==========");
    tr1.expense(101);
    System.out.println("6 ==========");
    tr1.expense(200);
    System.out.println("7 ==========");
    tr1.income(200);
    tr1.showHistory();
    System.out.println("8 ==========");
 }
}
```

Output

```
A new money tracker has been launched.
Name: null
Current Balance: 0.0
1 ==========
Name: John
Current Balance: 1.0
2 ==========
Balance Updated!
Name: John
Current Balance: 1001.0
3 ==========
Balance Updated.
Balance Updated.
Name: John
Current Balance: 101.0
4 ==========
Last added: 1000.0
Last spent: 100.0
5 ==========
You're broke!
6 ==========
Not enough balance.
7 ==========
Balance Updated!
Last added: 200.0
Last spent: 101.0
8 ==========
```

Task 5

```
public class A{
    public int x, y, z = 5;
    public double p = 0.0;
    public void methodA(int x, int m) {
        this.x = methodB(this.x);
        p = x + this.x % m * 3.0;
        y = y + methodB(x++, this.x);
        System.out.println(this.x +" " + x + y + " " + p) ;
    }
    public int methodB(int q, int n) {
        int arr[] = {3,4,5};
        arr[0] = arr[0] + this.x + n;
        arr[1] = q + arr[1];
        System.out.println(arr[0] +" " + arr[1] + " " + arr[2]) ;
        return arr[1] + arr[2];
    }
    public int methodB(int y) {
        if(y % 2 == 0) {
            int temp = this.methodB(2, y);
            return temp;
        }
        else{
            return 4;
       }
    }
}
```

Driver Code	Output		

`public class Tester11 {`
`  public static void main(String [] args){`
`  A a1 = new A();`
`  a1.methodA(2,3);`
`  a1.methodB(5,4);`
`  }`
`}`			

Task 6

```
1  public class Maze{
2          public int x;
3          public Maze(){
4            int x = 17;
5            this.methodB(x, -this.x);
6          }
7          public void methodA(){
8              int m = 0, x = 9;
9              m = methodB(m-3)+x;
10             this.x = ++x;
11             System.out.println(this.x+" "+m);
12             this.methodB(x,m);
13             System.out.println(x+" "+(m+this.x));
14             methodB(m);
15         }
16         public int methodB(int y){
17             x=y*y;
18             System.out.println(x+" "+y);
19             return x-11;
20         }
21         public void methodB(int z, int x){
22             z=z-2;
23             x=this.x-2*x;
24             System.out.println(z+" "+this.x);
25         }
26 }
```

DRIVER CODE	OUTPUTS	
public class MazeTester{		
public static void main(String args []){		
Maze m1 = new Maze();		
m1.methodA();		
new Maze().methodB(7,15);		
}		
}		

Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

Task 1

Driver Code	Output

`public class CustomerTester {`
`  public static void main(String[] args) {`
`  Customer c1 = new Customer();`
`  c1.createCustomer("John");`
`  System.out.println("1====================");`
`  c1.showCart();`
`  System.out.println("2====================");`
`  c1.addItem("Apple", 2);`
`  c1.addItem("Bread", 5);`
`  c1.addItem("Milk");`
`  c1.addItem("Eggs", 2);`
`  System.out.println("3====================");`
`  c1.showCart();`
`  System.out.println("4====================");`
`  c1.calculatePrice();`
`  System.out.println("5====================");`
`  Customer c2 = new Customer();`
`  c2.createCustomer("Jane");`
`  c2.addItem("Apple", 2, "Orange", 5);`
`  c2.addItem("Chocolates", 15, "Bread", 5);`
`  c2.addItem("Milk", 3);`
`  System.out.println("6====================");`
`  c2.showCart();`
`  System.out.println("7====================");`
`  c2.calculatePrice();`
`  }`
`}`
	`1====================`
	`Customer: John`
	`2====================`
	`Apple added to cart`
	`Orange added to cart`
	`Bread added to cart`
	`Milk added to cart`
	`Cart is full`
	`3====================`
	`Customer: John`
	`Item: Apple Price: 2`
	`Item: Orange Price: 5`
	`Item: Bread Price: 5`
	`Item: Milk Price: 3`
	`4====================`
	`Total: 15`
	`5====================`
	`Apple and Orange added to cart`
	`Chocolates and Bread added to cart`
	`Cart is full`
	`6====================`
	`Customer: Jane`
	`Item: Apple Price: 2`
	`Item: Orange Price: 5`
	`Item: Chocolates Price: 15`
	`Item: Bread Price: 5`
	`7====================`
	`Total: 27`

Task 2

Driver Code	Sample Output

`public class CalculatorTester {`
`  public static void main(String[] args) {`
`  Calculator calc = new Calculator();`
 	
`  System.out.println("1==========");`
`  calc.add(10, 20);`
`  System.out.println("2==========");`
`  calc.add(5, 15, 25);`
`  System.out.println("3==========");`
`  calc.multiply(6, 7);`
`  System.out.println("4==========");`
`  calc.multiply(2, 3, 4);`
`  System.out.println("5==========");`
`  calc.multiply("Hello", 3);`
`  System.out.println("6==========");`
`  calc.multiply("Java", 5);`
`  }`
`}`
	`1==========`
	`30`
	`2==========`
	`45`
	`3==========`
	`42`
	`4==========`
	`24`
	`5==========`
	`Hello-Hello-Hello`
	`6==========`
	`Java-Java-Java-Java-Java`

Task 3

Driver Code	Sample Output

`public class LibraryTest {`
`  public static void main(String[] args) {`
 	
`  Book book1 = new Book();`
`  book1.createBook("The Great Gatsby");`
 	
`  Book book2 = new Book();`
`  book2.createBook("1984", "George Orwell");`
 	
`  Book book3 = new Book();`
`  book3.createBook("To Kill a Mockingbird", "Harper Lee", "Fiction");`
 	
`  System.out.println("|---Book Customization---|");`
`  book1.customizeGenre("Classic");`
`  book1.customizePages(180);`
 	
`  book2.customizeGenre("Dystopian");`
`  book2.customizePages(328);`
 	
`  book3.customizePages(281);`
 	
`  System.out.println();`
 	
`  System.out.println("|---Library Inventory---|");`
`  book1.displayDetails();`
`  book2.displayDetails();`
`  book3.displayDetails();`
`  }`
`}`
	`|---Book Customization---|`
	`Updated genre of "The Great Gatsby" to Classic.`
	`Updated pages of "The Great Gatsby" to 180 pages.`
	`Updated genre of "1984" to Dystopian.`
	`Updated pages of "1984" to 328 pages.`
	`Updated pages of "To Kill a Mockingbird" to 281 pages.`
	
	`|---Library Inventory---|`
	`Title: The Great Gatsby, Author: Unknown, Genre: Classic, Pages: 180`
	`Title: 1984, Author: George Orwell, Genre: Dystopian, Pages: 328`
	`Title: To Kill a Mockingbird, Author: Harper Lee, Genre: Fiction, Pages: 281`

Task 4

Driver Code	Sample Output

`public class MovieManagerTest {`
`  public static void main(String[] args) {`
`  Movie inception = new Movie();`
`  inception.setMovieDetails("Inception", "Christopher Nolan", 8.8);`
 	
`  System.out.println("1====================");`
`  inception.addActors("Leonardo DiCaprio", "Joseph Gordon-Levitt");`
`  inception.addActors("Ellen Page");`
`  inception.showInfo();`
 	
`  System.out.println("2====================");`
`  Movie avengers = new Movie();`
`  avengers.setMovieDetails("Avengers: Endgame", "Anthony Russo", 8.4);`
`  avengers.addActors("Robert Downey Jr.", "Chris Evans", "Scarlett Johansson");`
`  avengers.showInfo();`
 	
`  System.out.println("3====================");`
`  Movie parasite = new Movie();`
`  parasite.setMovieDetails("Parasite", "Bong Joon-ho");`
`  parasite.addActors("Song Kang-ho", "Choi Woo-shik");`
`  parasite.updateRating(8.6);`
`  parasite.showInfo();`
 	
`  System.out.println("4====================");`
`  parasite.updateRating(8.9);`
`  parasite.showInfo();`
`  }`
`}`
	`1====================`
	`Added actor "Leonardo DiCaprio" to "Inception".`
	`Added actor "Joseph Gordon-Levitt" to "Inception".`
	`Added actor "Ellen Page" to "Inception".`
	`Title: Inception`
	`Director: Christopher Nolan`
	`Rating: 8.8`
	`Actors: Leonardo DiCaprio, Joseph Gordon-Levitt, Ellen Page`
	`2====================`
	`Added actor "Robert Downey Jr." to "Avengers: Endgame".`
	`Added actor "Chris Evans" to "Avengers: Endgame".`
	`Added actor "Scarlett Johansson" to "Avengers: Endgame".`
	`Title: Avengers: Endgame`
	`Director: Anthony Russo`
	`Rating: 8.4`
	`Actors: Robert Downey Jr., Chris Evans, Scarlett Johansson`
	`3====================`
	`Added actor "Song Kang-ho" to "Parasite".`
	`Added actor "Choi Woo-shik" to "Parasite".`
	`Updated rating of "Parasite" to 8.6`
	`Title: Parasite`
	`Director: Bong Joon-ho`
	`Rating: 8.6`
	`Actors: Song Kang-ho, Choi Woo-shik`
	`4====================`
	`Updated rating of "Parasite" to 8.9`
	`Title: Parasite`
	`Director: Bong Joon-ho`
	`Rating: 8.9`
	`Actors: Song Kang-ho, Choi Woo-shik`

Task 5

```
public class Test3 {
    int x = 2, y = 4, sum = 3;
    int arr[] = {x, y, sum};
    public void methodA(int x) {
        arr[0] += methodB(y, this.x) + methodC(x);
        System.out.println(x + " " + this.x + " " + sum);
        arr[1] += this.x * (++y) / (sum % x);
        System.out.println(y + " " + sum + " " + this.x);
        arr[2] += methodC(x) + methodB(this.x, sum);
        System.out.println(arr[0] + " " + arr[1] + " " + arr[2]);
    }
    public int methodB(int q, int n) {
        int arr2[] = {7, 8};
        int a = (arr2[0]++) - q;
        int b = (++arr2[1]) - n;
        return a + b;
    }
    public int methodC(int z) {
        z = sum + methodB(x, sum) - z;
        return z/2;
    }
}
```

```
public class Tester3{
    public static void main(String [] args){
        Test3 t3 = new Test3();
        t3.methodA(7);
    }
}
```

Outputs		